

The Animal Kingdom Express

I PUC · Ch 4 · Animal Kingdom

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



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All Aboard!

"All aboard the Animal Express!" called Prof. BAIO. "Each coach is a phylum. Tickets are checked by **levels of organisation**, **symmetry**, **germ layers** and the **coelom** — the body cavity."

💡 **REMEMBER**

Basis of classification: level of organisation, symmetry, diploblastic/triploblastic, coelom, segmentation and notochord.



Coach 1 · The Sponges

Sponges (**Porifera**) sat still, covered in holes. Water entered through tiny **ostia**, swirled through the **spongocoel** and left through the **osculum**. Collar cells called **choanocytes** lined the canals, catching food.

💡 **REMEMBER**

Porifera: cellular level of organisation, water canal system, mostly marine.



Coach 2 · The Stingers

Cnidarians waved tentacles armed with **cnidoblasts** — stinging cells. Some were **polyp** shaped like Hydra; others were umbrella-shaped **medusa** like Aurelia. Their cousins, the comb jellies (**Ctenophora**), glowed in the dark — **bioluminescence**!

💡 **REMEMBER**

Cnidoblasts help in anchorage, defence and catching prey. Ctenophores have eight rows of ciliated comb plates.



Coach 3 · The Worms

Flatworms (**Platyhelminthes**) like the tapeworm excreted with **flame cells**. Roundworms (**Aschelminthes**) like Ascaris had a false body cavity, the **pseudocoelom**. Earthworms (**Annelida**) had a true coelom, a body of repeated segments (**metamerism**) and **nephridia**.

💡 **REMEMBER**

Platyhelminthes — acoelomate · Aschelminthes — pseudocoelomate · Annelida — coelomate and segmented.



Coach 4 · Jointed Legs

The busiest coach of all! **Arthropoda** — the largest phylum, with two-thirds of all named species. They wore a **chitin** exoskeleton, walked on jointed legs and excreted through **Malpighian tubules**. Bees, mosquitoes, locusts and the living fossil Limulus all had seats here.

💡 **REMEMBER**

Arthropoda: jointed appendages, chitinous exoskeleton, open circulation, Malpighian tubules.



Coach 5 · Shells and Spines

Molluscs like Pila scraped food with a file-like **radula**. Starfish (**Echinodermata**) moved with a **water vascular system**; the adults were radially symmetrical, but their larvae were bilateral. In a corner, the worm-like **Balanoglossus** (a hemichordate) waved hello.

💡 **REMEMBER**

Water vascular system — locomotion, food capture and respiration in echinoderms.



The Chordate Check

At the last station, three questions decided who was a **chordate**: Do you have a **notochord**? A dorsal, hollow **nerve cord**? **Pharyngeal gill slits**? Fishes, frogs, reptiles, birds and mammals all shouted, "Yes!"

💡 **REMEMBER**

Non-chordates have no notochord, a ventral solid nerve cord and no gill slits; their heart is dorsal.



The Vertebrate Family

Prof. BAIO introduced the classes. Amphibia could live on land and in water. Reptilia had dry, scaly skin. **Aves** had feathers and hollow bones. **Mammalia** had **mammary glands**, hair and external ears. The platypus giggled — "I am a mammal that lays eggs!"

💡 REMEMBER

Mammals: mammary glands, hair, external pinnae; mostly viviparous (platypus lays eggs).



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

levels of organisation p.1 · symmetry p.1 · germ layers p.1 · coelom p.1

Porifera p.2 · ostia p.2 · spongocoel p.2 · osculum p.2

choanocytes p.2 · cnidoblasts p.3 · polyp p.3 · medusa p.3

Ctenophora p.3 · bioluminescence p.3 · Platyhelminthes p.4

flame cells p.4 · Aschelminthes p.4 · pseudocoelom p.4 · Annelida p.4

metamerism p.4 · nephridia p.4 · Arthropoda p.5 · chitin p.5

Malpighian tubules p.5 · radula p.6 · Echinodermata p.6

water vascular system p.6 · Balanoglossus p.6 · chordate p.7

notochord p.7 · nerve cord p.7 · Pharyngeal gill slits p.7 · Aves p.8

Mammalia p.8 · mammary glands p.8



QUICK QUIZ

Do you remember?

1. Water vascular system is found in:

- Mollusca
- Echinodermata
- Annelida
- Porifera

2. The largest animal phylum is:

- Chordata
- Mollusca
- Arthropoda
- Annelida

3. Flame cells are excretory cells of:

- Platyhelminthes
- Annelida
- Arthropoda
- Cnidaria

4. The radula is found in:

- Mollusca
- Echinodermata
- Porifera
- Cnidaria

5. Which is NOT a chordate feature?

- Notochord
- Dorsal hollow nerve cord
- Pharyngeal gill slits
- Ventral solid nerve cord

Answers: 1-Echinodermata · 2-Arthropoda · 3-Platyhelminthes · 4-Mollusca · 5-Ventral solid nerve cord